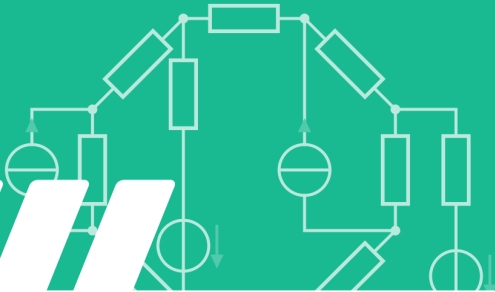


GET it digital

Modul 5: Extended DC networks



Stand: 18. Februar 2026



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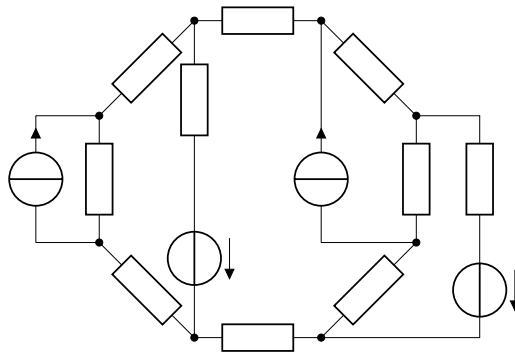
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<https://getitdigital.uni-wuppertal.de/module/modul-5-erweiterte-gleichstromnetzwerke>

The “Advanced DC Networks” module covers the following topics:

- ▶ Node and mesh rule
- ▶ Superposition
- ▶ Node potential analysis
- ▶ Mesh current analysis



The analysis of nodes and meshes in electrical networks is carried out using Kirchhoff's rules.

Learning objectives: Knot- and Mesh analysis

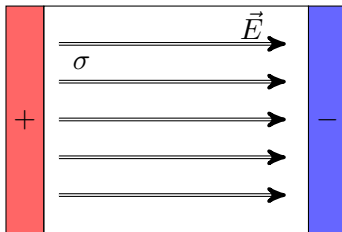
The Students

1. know the basic definitions for describing an electrical network.
2. can apply the node rule and the mesh rule to electrical networks.

Current density in free space

Electrical Field $\vec{E}$ between two charged plates:

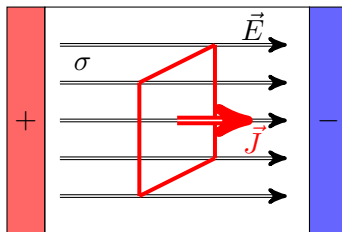
- The space between the plates is filled with a conductive material σ .



Current density in free space

Electrical Field $\vec{E}$ between two charged plates:

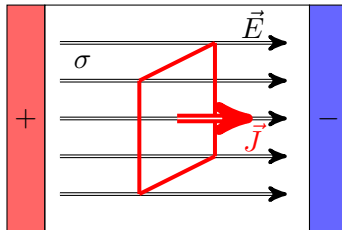
- ▶ The space between the plates is filled with a conductive material σ .
- ▶ Current density $\vec{J}$ in the electric field



Current density in free space

Electrical Field $\vec{E}$ between two charged plates:

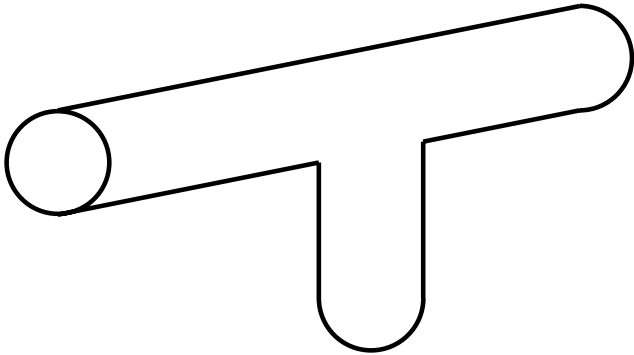
- ▶ The space between the plates is filled with a conductive material σ .
- ▶ Current density $\vec{J}$ in the electric field
- ▶ Source-free current density (continuity equation)



$$\oint \vec{J} d\vec{A} = 0 \quad \rightarrow \quad \text{div} \vec{J} = 0$$

Current density in structured space

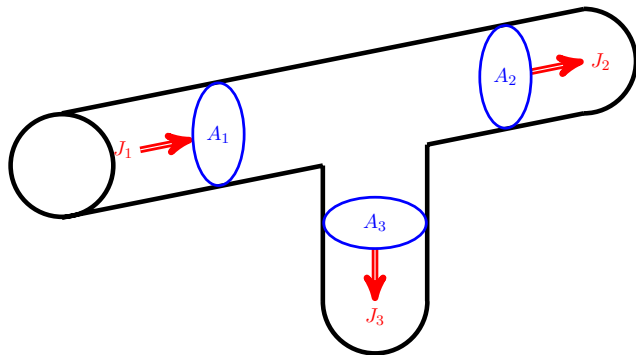
Defined field space:



Current density in structured space

Defined field space:

- Known areas and current densities

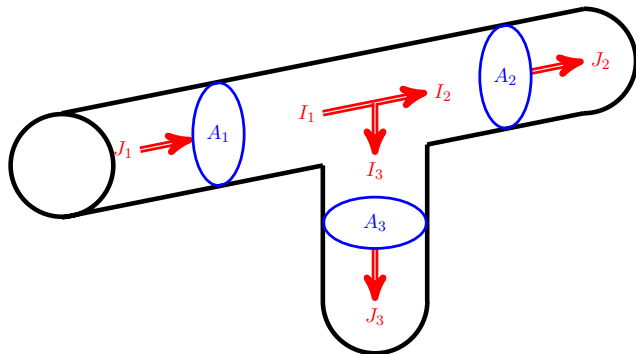


Current density in structured space

Defined field space:

- ▶ Known areas and current densities
- ▶ Current

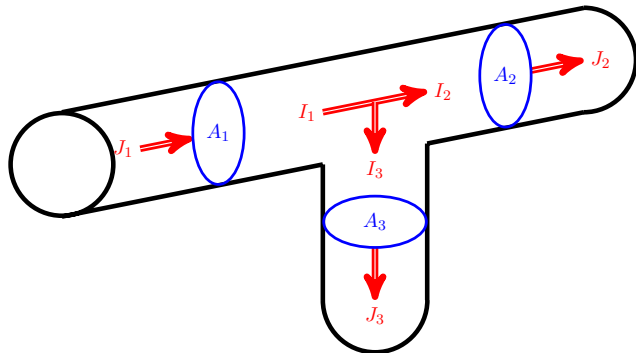
$$I_1 = \iint_A \vec{J}_1 d\vec{A}$$



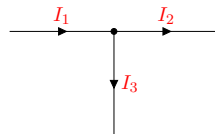
Current density in structured space

Defined field space:

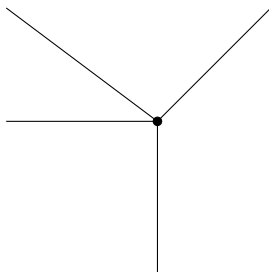
- ▶ Known areas and current densities
- ▶ Current
- ▶ Network



$$I_1 = \iint_A \vec{J}_1 d\vec{A}$$

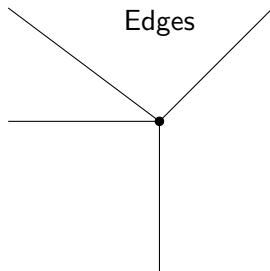


Relationship between nodes and branches in graph theory:



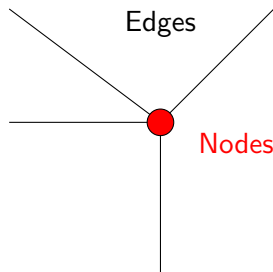
Relationship between nodes and branches in graph theory:

- Edges $\rightarrow$ Branches



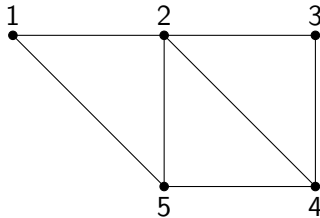
Relationship between nodes and branches in graph theory:

- ▶ Edges $\rightarrow$ Branches
- ▶ Knots



Digression: Graph theory

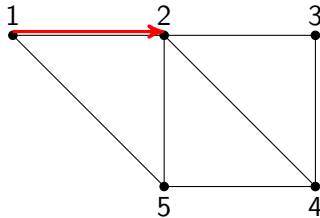
Graph:



Digression: Graph theory

Graph:

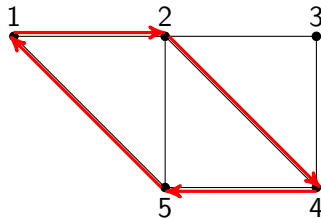
► Path



Digression: Graph theory

Graph:

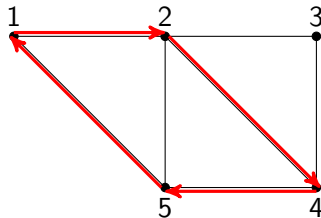
- ▶ Path
- ▶ Start of path = End of path $\rightarrow$ Cycle



Digression: Graph theory

Graph:

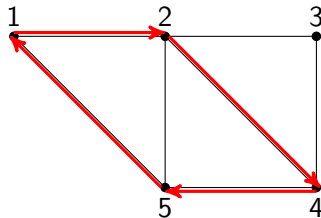
- ▶ Path
- ▶ Start of path = End of path $\rightarrow$ Cycle
- ▶ No crossing cycle $\rightarrow$ circle



Digression: Graph theory

Graph:

- ▶ Path
- ▶ Start of path = End of path → Cycle
- ▶ No crossing cycle → circle

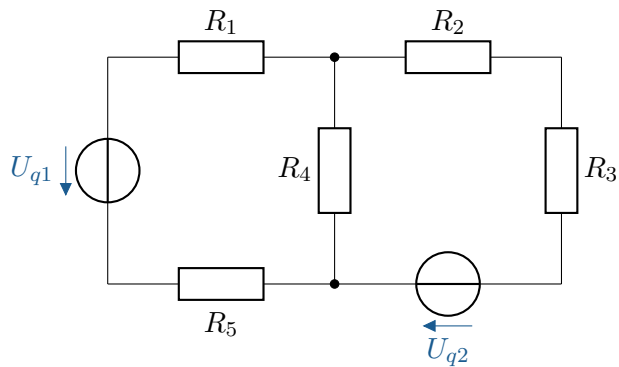


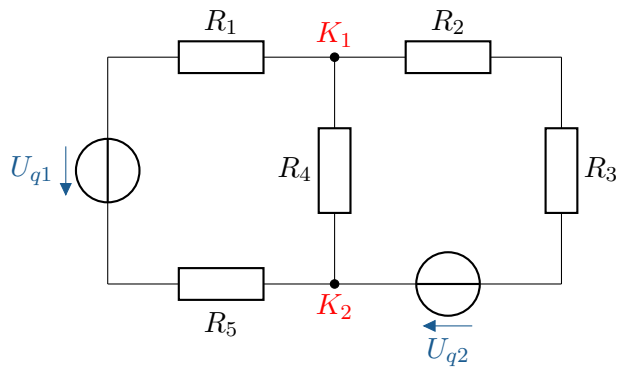
Associated adjacency matrix:

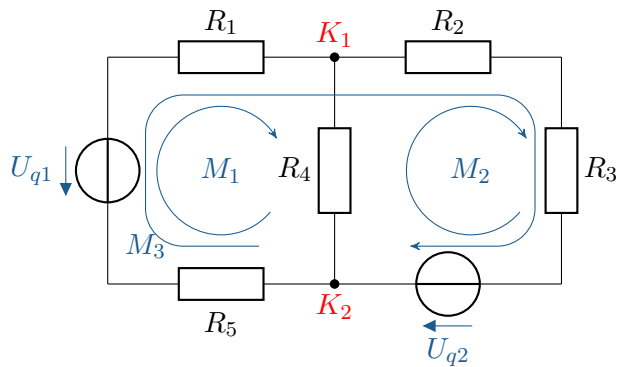
$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \end{bmatrix}$$

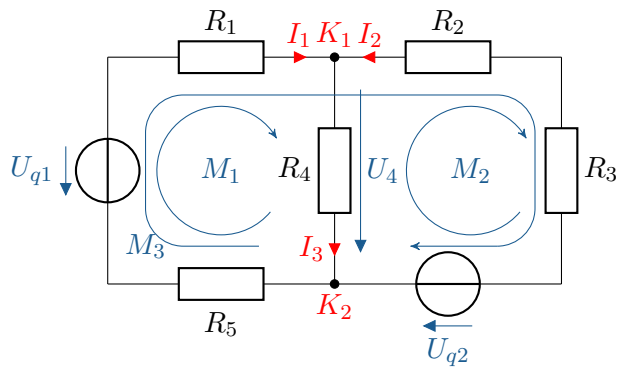
Key point: Graph theory

In graph theory, networks are described using edges and nodes. When analysing networks, paths and cycles are defined. A continuous cycle with an identical starting point and end point is referred to as a circle.





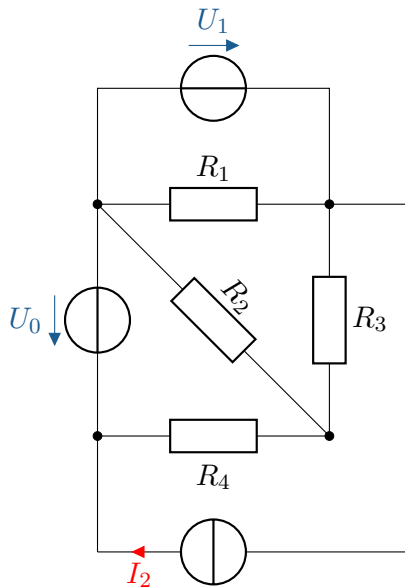




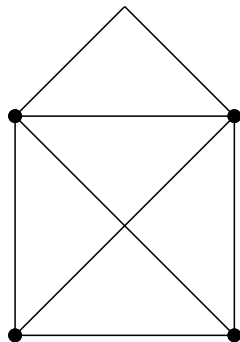
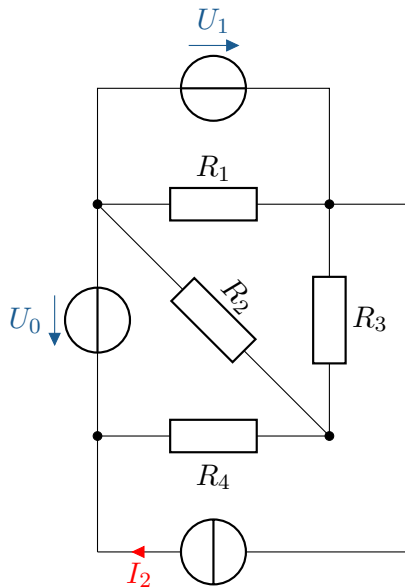
Key point: Knots, Branches and Meshes

Electrical networks are described by nodes, branches and meshes, analogous to graph theory.

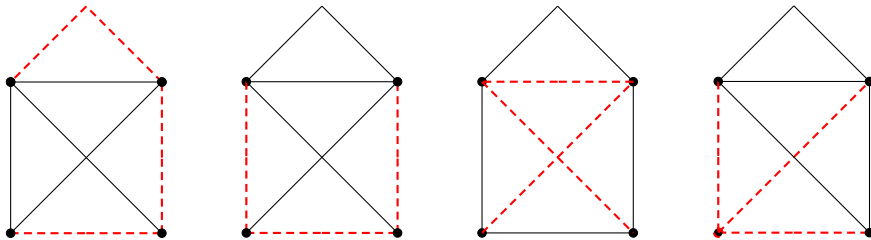
The complete tree



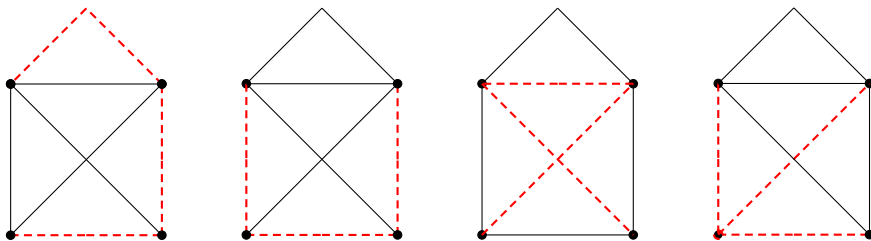
The complete tree



The complete tree

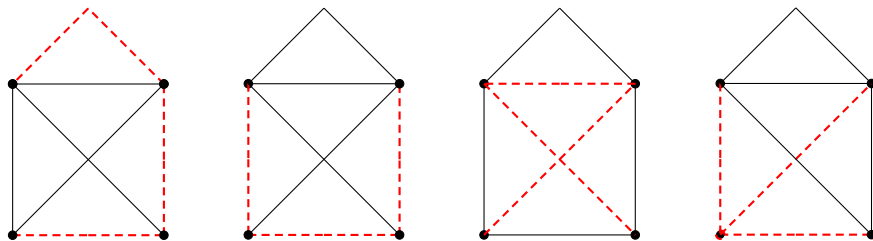


The complete tree



- ▶ **Tree branches** → $k-1$ of the complete tree
- ▶ the remaining branches are connecting branches

The complete tree



- ▶ **Tree branches** $\rightarrow k-1$ of the complete tree
- ▶ the remaining branches are connecting branches
- ▶ A network with z branches and k nodes contains $z - k + 1$ connecting branches.

Key point: The complete tree

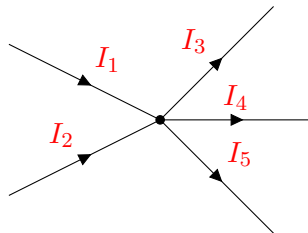
With the help of a complete tree, all nodes are connected to each other by tree branches. This always results in $k - 1$ tree branches. The complete tree does not form a mesh.

Knot rule

- ▶ The sum of all currents at a node is equal to zero.

$$\sum_{k=1}^N I_k = 0$$

- ▶ Example network:



- ▶ Sum of inflows and outflows equals zero:

$$\sum_{k=1}^N I_k = 0 = I_1 + I_2 - I_3 - I_4 - I_5$$

Key point: Knot rule

The sum of the incoming flows is equal to the sum of the outgoing flows, or the sum of all flows at a node is zero:

$$\sum_{k=1}^N I_k = 0$$

Ring integral over the field strength:

- Condition: No time-varying magnetic flux density and therefore no electric eddy field.

$$\oint_s \vec{E} d\vec{s} = 0 \quad \rightarrow \quad \text{rot} \vec{E} = 0$$

Ring integral over the field strength:

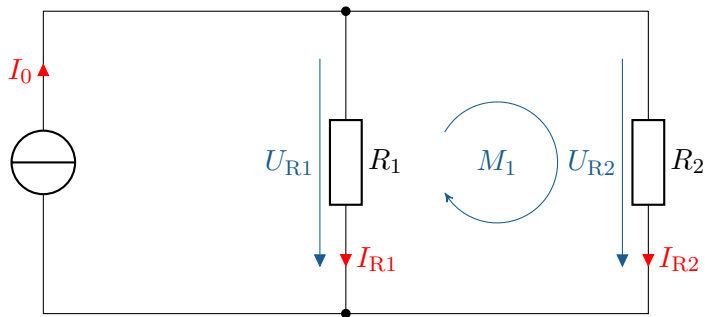
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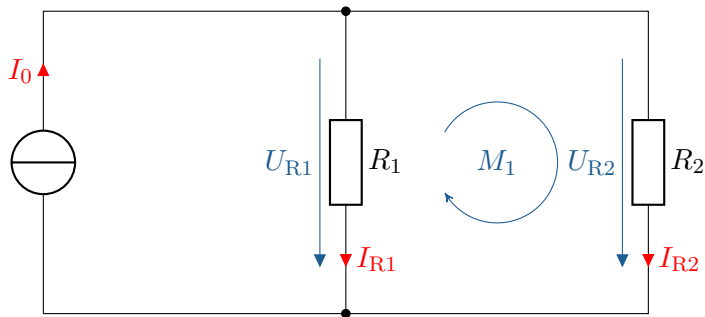
$$\oint_s \vec{E} d\vec{s} = 0 \quad \rightarrow \quad \text{rot} \vec{E} = 0$$

Mesh rule:

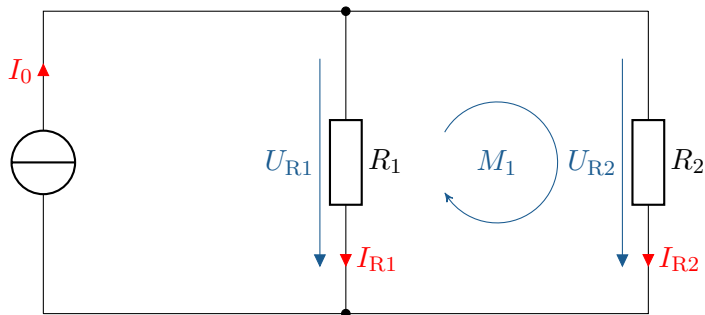
- In a complete circuit in an electrical network, the sum of all voltages is equal to zero.

$$\sum_{k=1}^N U_k = 0$$

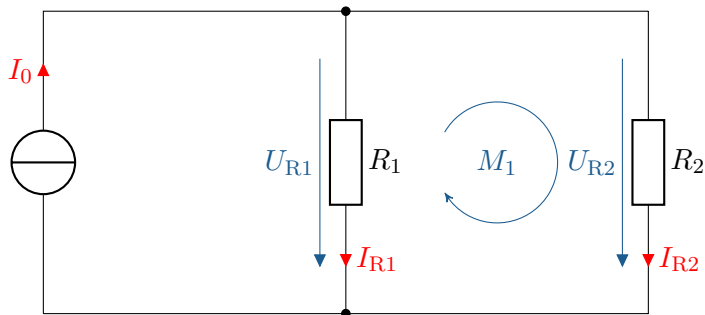




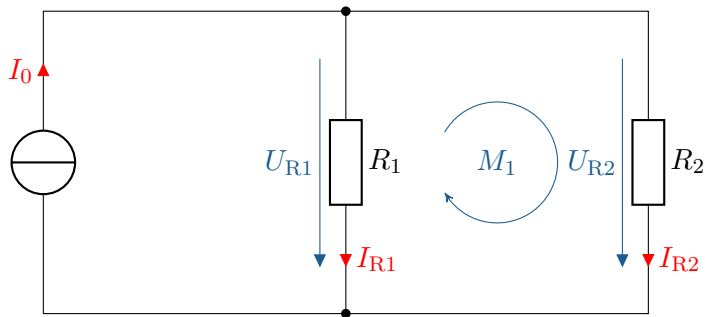
$$\sum_{k=1}^N U_k$$



$$\sum_{k=1}^N U_k = +U_{R2}$$



$$\sum_{k=1}^N U_k = +U_{R2} - U_{R1}$$



$$\sum_{k=1}^N U_k = +U_{R2} - U_{R1} = 0$$

Key point: Mesh rule

In a mesh, the sum of all partial stresses is equal to zero.:

$$\sum_{k=1}^N U_k = 0$$

Objective: Calculation of a complete network. Steps of a procedure:

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1. Simplifying the network.

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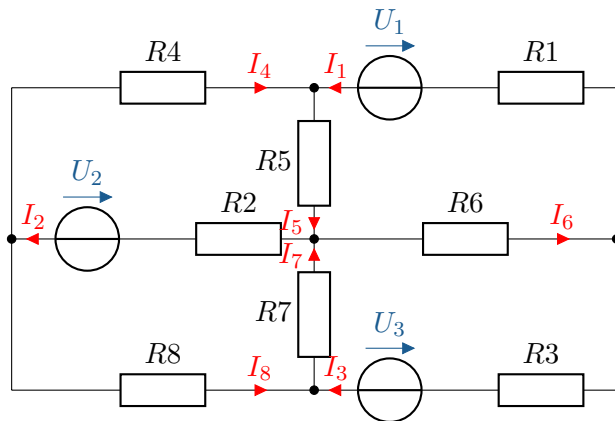
1. Simplifying the network.
2. Draw all current arrows and create the graph.
3. Set up $k - 1$ node equations.
4. Set up the $m = z - k + 1$ linearly independent mesh equations.

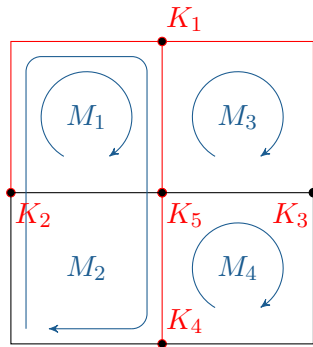
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5. Solving the system of equations with the dimension z .

Objective: Calculation of a complete network. Steps of a procedure:

1. Simplifying the network.
2. Draw all current arrows and create the graph.
3. Set up $k - 1$ node equations.
4. Set up the $m = z - k + 1$ linearly independent mesh equations.
5. Solving the system of equations with the dimension z .
6. If necessary, undo step 1.





$$K_1 : \quad I_1 + I_4 - I_5 = 0$$

$$K_2 : \quad I_2 - I_4 - I_8 = 0$$

$$K_3 : \quad -I_1 - I_3 + I_6 = 0$$

$$K_4 : \quad I_3 - I_7 + I_8 = 0$$

$$M_1 : \quad R_2 I_2 + R_4 I_4 + R_5 I_5 - U_2 = 0$$

$$M_2 : \quad R_4 I_4 + R_5 I_5 - R_7 I_7 - R_8 I_8 = 0$$

$$M_3 : \quad -R_1 I_1 - R_5 I_5 - R_6 I_6 + U_1 = 0$$

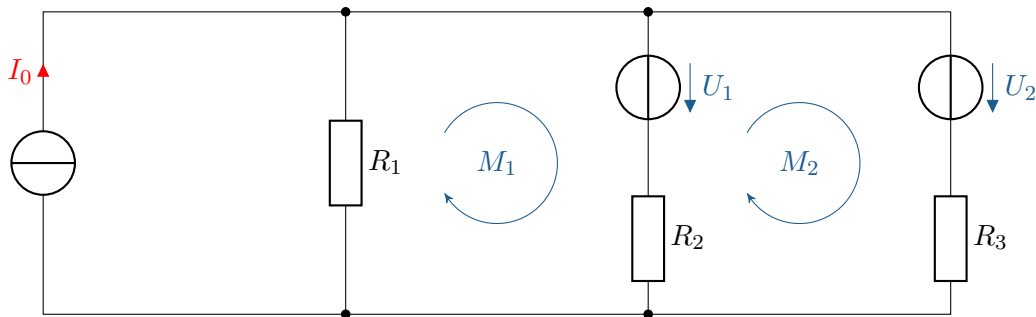
$$M_4 : \quad R_3 I_3 + R_6 I_6 + R_7 I_7 - U_3 = 0$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & 1 \\ 0 & R_2 & 0 & R_4 & R_5 & 0 & 0 & 0 \\ 0 & 0 & 0 & R_4 & R_5 & 0 & -R_7 & -R_8 \\ -R_1 & 0 & 0 & 0 & -R_5 & -R_6 & 0 & 0 \\ 0 & 0 & R_3 & 0 & 0 & R_6 & R_7 & 0 \end{bmatrix} \cdot \begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \\ I_6 \\ I_7 \\ I_8 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ U_2 \\ 0 \\ -U_1 \\ U_3 \end{pmatrix}$$

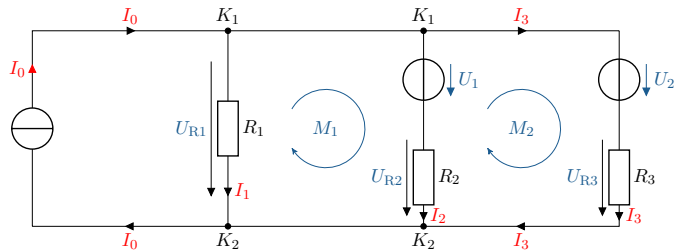
Example network analysis

The specified electrical network is given. The following tasks are to be completed:

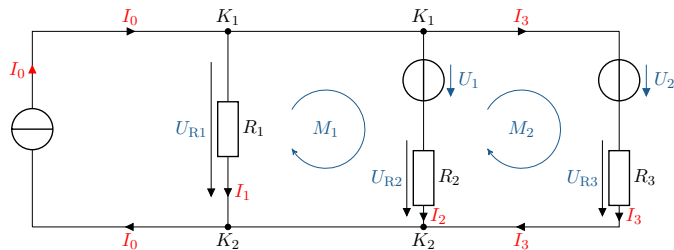
- a) Plotting currents and voltages
- b) Setting up the node equations
- c) Setting up the mesh equation M_1
- d) Setting up the mesh equation M_2 without the expression U_1



a) Specification of currents and voltages:



Example network analysis

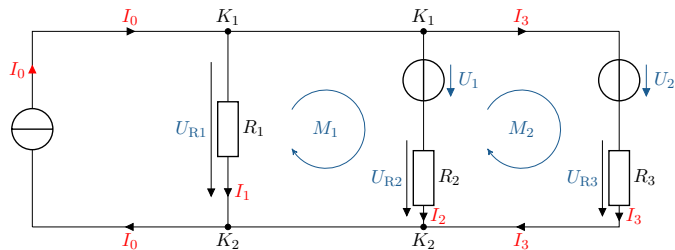


b) Knot equations:

$$K_1 : I_0 - I_1 - I_2 - I_3 = 0$$

$$K_2 : -I_0 + I_1 + I_2 + I_3 = 0$$

Example network analysis

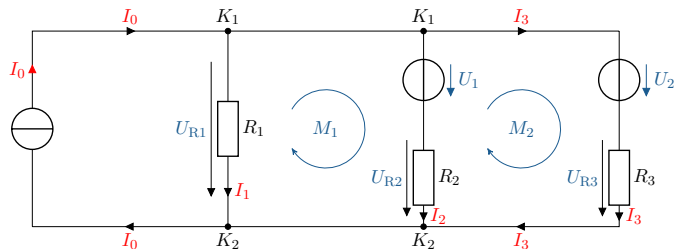


c) Setting up the mesh equation M_1 and M_2 :

$$M_1 : U_1 + U_{R2} - U_{R1} = 0$$

$$M_2 : U_2 + U_{R3} - U_{R2} - U_1 = 0$$

Example network analysis



d) Setting up the mesh equation M_2 without the expression U_1

$$M_1 : U_1 = U_{R1} - U_{R2}$$

$$M_2 : U_2 + U_{R3} - U_{R2} - (U_{R1} - U_{R2}) = 0$$

$$M_2 : U_2 + U_{R3} - \cancel{U_{R2}} - U_{R1} + \cancel{U_{R2}} = 0$$

$$M_2 : U_2 + U_{R3} - U_{R1} = 0$$

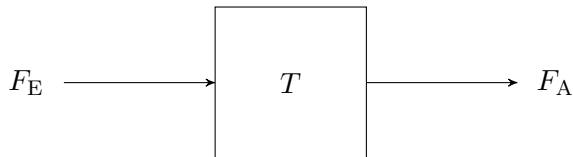


Learning objectives: Principle of superposition

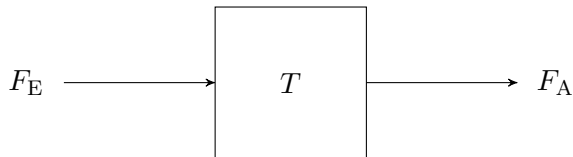
The students

- ▶ understand the conditions of systems theory for the analysis of direct current networks.
- ▶ can apply the superposition principle to electrical networks.

- Consideration of a black box as a system with input variables and output variables.



- Consideration of a black box as a system with input variables and output variables.

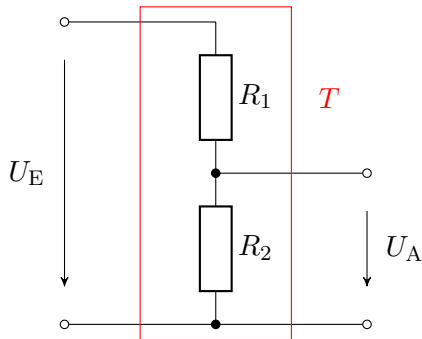


- Transformation of the input signal F_E into an output signal F_A .

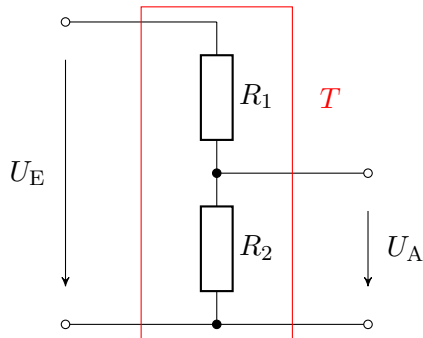
$$F_A = T(F_E)$$

- ▶ Description of a black box as an electrical network with an input voltage and an output voltage.

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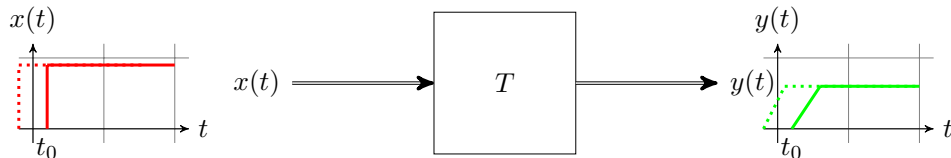


- Transformation of the input voltage U_E into an output voltage U_A .

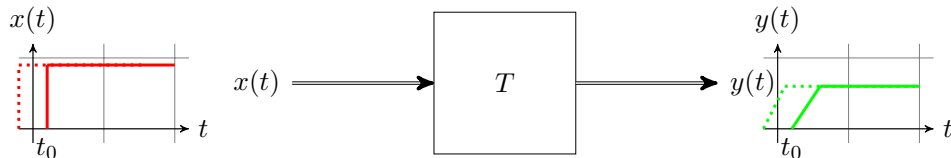
$$U_A = T(U_E) = U_E \cdot \frac{R_2}{R_1 + R_2}$$

- ▶ Causal systems do not exhibit a system response prior to excitation.

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- ▶ Time-invariant systems always respond identically to an input signal, regardless of the timing.
- ▶ Linear system $\rightarrow$ finite output amplitude for finite input amplitude



- ▶ Causal systems do not exhibit a system response prior to excitation.
- ▶ Time-invariant systems always respond identically to an input signal, regardless of the timing.
- ▶ Linear system $\rightarrow$ finite output amplitude for finite input amplitude
- ▶ Time-invariant and linear systems are categorised as LTI systems.



Key point: LTI-Systems

LTI systems represent linear and time-invariant systems. DC networks must be considered as LTI systems so that the superposition theorem can be applied to them.

Superposition theorem

Conditions and execution of the overlay set:

- ▶ Assumed linearity

Superposition theorem

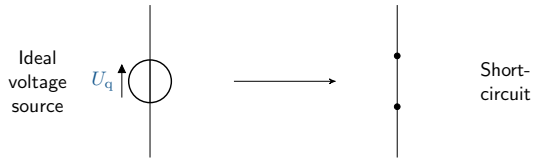
Conditions and execution of the overlay set:

- ▶ Assumed linearity
- ▶ Switching off the sources

Superposition theorem

Conditions and execution of the overlay set:

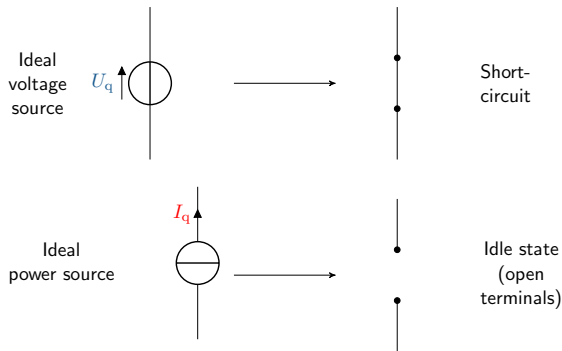
- ▶ Assumed linearity
- ▶ Switching off the sources
- ▶ Ideal voltage source replaced by a short circuit



Superposition theorem

Conditions and execution of the overlay set:

- ▶ Assumed linearity
- ▶ Switching off the sources
- ▶ Ideal voltage source replaced by a short circuit
- ▶ Ideal power source is replaced by open terminals, an open circuit



Key point: Conversion of sources

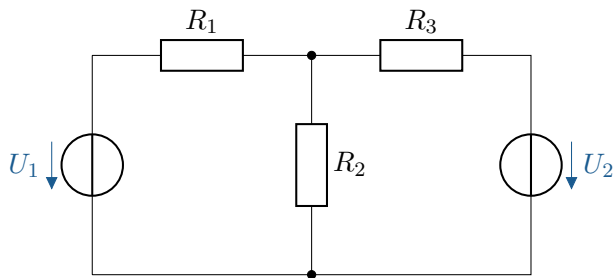
When voltage sources and current sources are switched off, deactivated voltage sources leave a short circuit and deactivated current sources leave an open circuit.

Assumption :

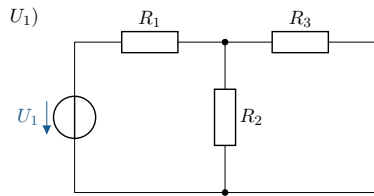
- ▶ The resistor R is supplied by the two sources Q_1 and Q_2 .
- ▶ Separate calculation of the current through R
- ▶ The total current through R is calculated from the sum of the currents from the two sources:

$$I_R = f(Q_1, Q_2)$$

Calculation of a sample network:

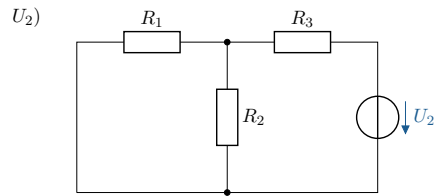
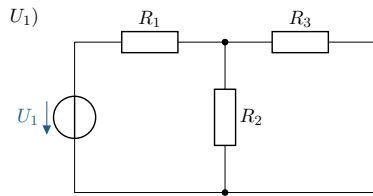


Example solution



$$U_{R3}(U_1) = U_1 \cdot \frac{R_{23}}{R_1 + R_{23}}$$

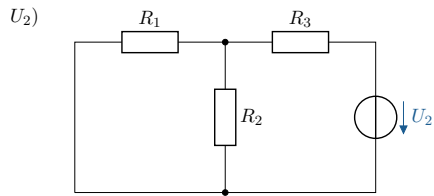
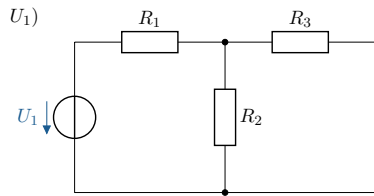
Example solution



$$U_{R3}(U_1) = U_1 \cdot \frac{R_{23}}{R_1 + R_{23}}$$

$$U_{R3}(U_2) = U_2 \cdot \frac{R_{12}}{R_3 + R_{12}}$$

Example solution



$$U_{R3}(U_1) = U_1 \cdot \frac{R_{23}}{R_1 + R_{23}}$$

$$U_{R3}(U_2) = U_2 \cdot \frac{R_{12}}{R_3 + R_{12}}$$

$$U_{R3} = U_{R3}(U_1) + U_{R3}(U_2)$$

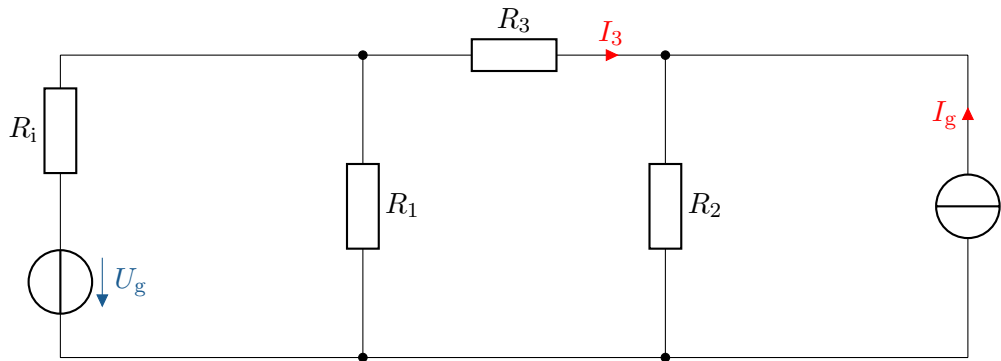
Key point: Superposition principle

Linear and time-invariant electrical networks with more than one source can be determined as the sum of the partial analyses of each individual source.

Example: Superposition method

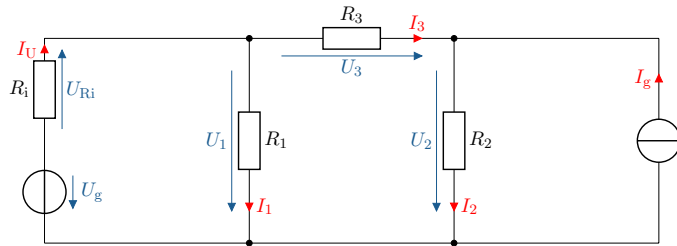
Application of the superposition principle (Superposition theorem):

- a) Plotting currents and voltages
- b) Network calculation I_3 for U_g
- c) Network calculation I_3 for I_g
- d) How big is the current? I_3



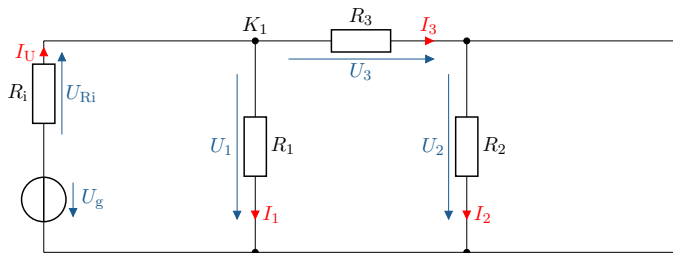
Example: Superposition method

a) Specification of currents and voltages:



Example: Superposition method

b) Network calculation $I_3(U_g)$ for U_g :



$$K_1 : I_{\text{tot}} = I_1 + I_3$$

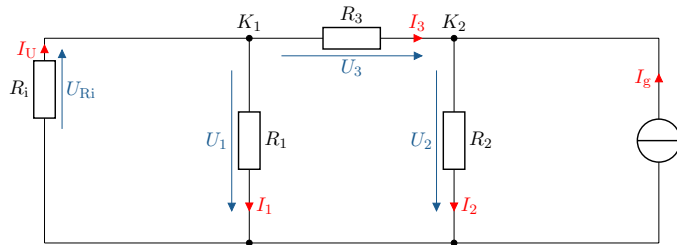
$$I_{\text{tot}} = \frac{U_g}{R_{\text{tot}}} = \frac{U_g}{R_i + \frac{R_1 \cdot (R_2 + R_3)}{R_1 + R_2 + R_3}}$$

$$\frac{I_3}{I_{\text{tot}}} = \frac{R_1}{R_1 + R_2 + R_3}$$

$$I_3(U_g) = I_{\text{tot}} \cdot \frac{R_1}{R_1 + R_2 + R_3} = \frac{U_g}{R_i + \frac{R_1 \cdot (R_2 + R_3)}{R_1 + R_2 + R_3}} \cdot \frac{R_1}{R_1 + R_2 + R_3}$$

Example: Superposition method

c) Network calculation $I_3(I_g)$ for I_g :



$$K_2 : I_{\text{tot}} = I_g = I_2 - I_3$$

$$\frac{-I_3}{I_{\text{tot}}} = \frac{R_2}{(R_1 || R_i) + R_2 + R_3}$$

$$I_3(I_g) = -I_g \cdot \frac{R_2}{(R_1 || R_i) + R_2 + R_3}$$

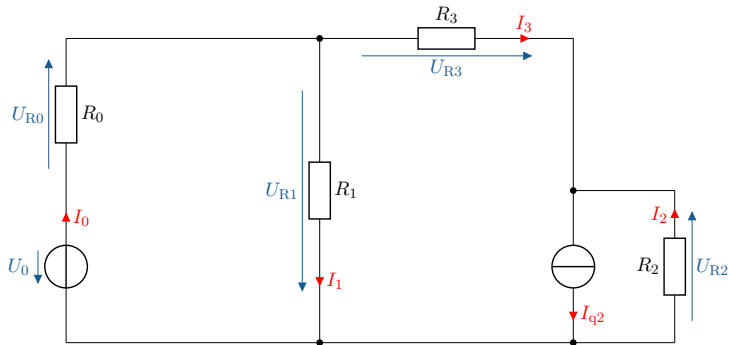
- d) How strong is the current $I_3(U_g, I_g)$?
Superposition:

$$I_3(U_g, I_g) = I_3(U_g) + I_3(I_g)$$

$$I_3(U_g, I_g) = \frac{U_g}{R_i + \frac{R_1 \cdot (R_2 + R_3)}{R_1 + R_2 + R_3}} \cdot \frac{R_1}{R_1 + R_2 + R_3} + (-I_g \cdot \frac{R_2}{(R_1 || R_i) + R_2 + R_3})$$

Nodal analysis

- ▶ Preparation of the network
- ▶ Determination of nodes and node potentials
- ▶ Allocation of source currents
- ▶ Setting up the conduct value matrix
- ▶ Setting up the system of equations



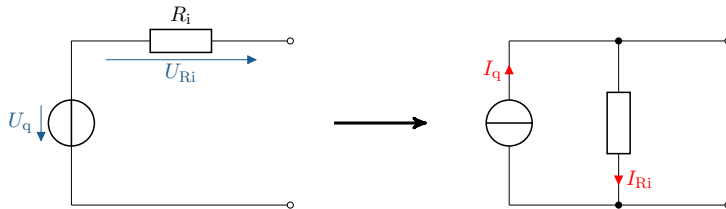
Learning objectives: Node analysis

The Students

- ▶ are familiar with the steps of the nodal analysis
- ▶ can analyse electrical networks using the nodal analysis method.

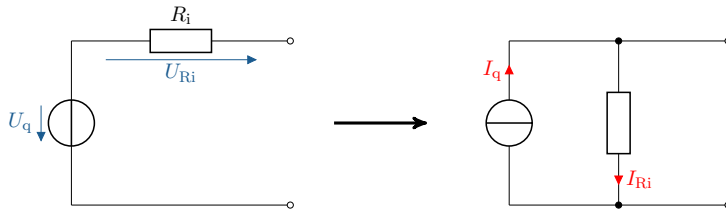
Preparation of the network

- Conversion of the voltage source into a current source:



Preparation of the network

- Conversion of the voltage source into a current source:



- Conversion of voltage source to current source:

$$I_q = U_q \cdot G_i$$

Preparation of the network

- Conversion of the voltage source into a current source:



- Conversion of voltage source to current source:

$$I_q = U_q \cdot G_i$$

- Calculation of the conductance values:

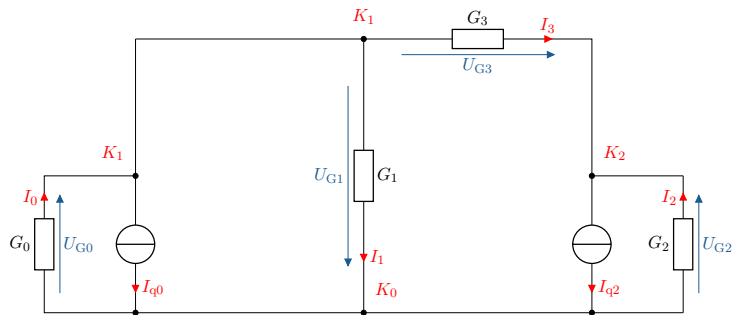
$$G_i = \frac{1}{R_i}$$

Key point: Nodal analysis

In the nodal analysis method, an electrical network with current sources and conductances is analysed.

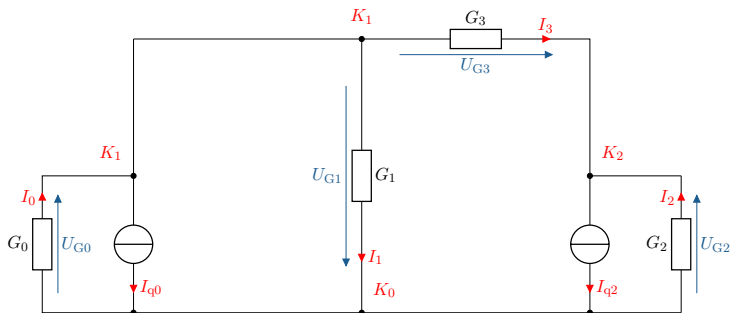
Determination of nodes and node potentials

- Determination of the reference node K_0 and the consecutive nodes K_1 and K_2 :



Determination of nodes and node potentials

- Determination of the reference node K_0 and the consecutive nodes K_1 and K_2 :



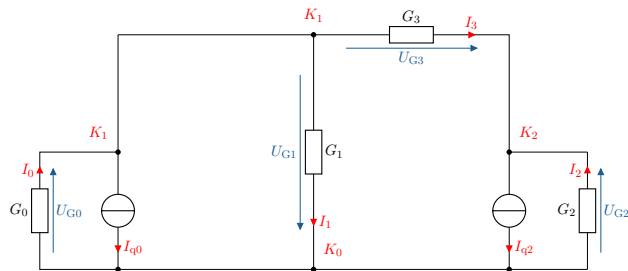
- $k - 1$ equations are required. This results in the following stress vector:

$$U_K = \begin{bmatrix} U_{K1-K0} \\ U_{K2-K0} \end{bmatrix}$$

- ▶ Assignment of source currents for each node
- ▶ Outflowing spring streams have a negative sign.
- ▶ Inflowing source streams are noted positively.

$$I_K = \begin{bmatrix} -I_{q0} \\ -I_{q2} \end{bmatrix}$$

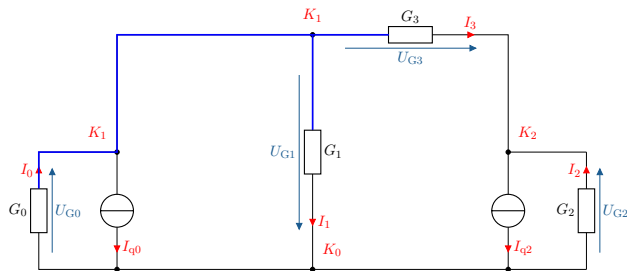
Conductance matrix



$$G_K = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Conductance matrix

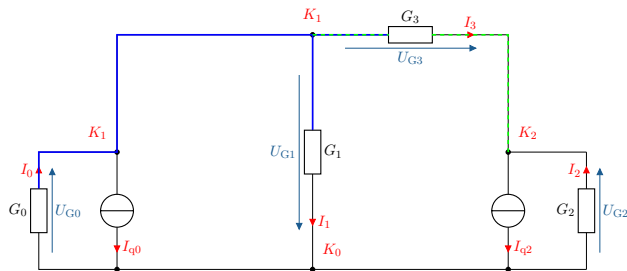
- Main diagonal with the conductance values directly adjacent to the nodes



$$G_K = \begin{bmatrix} G_0 + G_1 + G_3 & 0 \\ 0 & G_2 + G_3 \end{bmatrix}$$

Conductance matrix

- ▶ Main diagonal with the conductance values directly adjacent to the nodes
- ▶ Other elements with the conductance values that lie directly between the nodes



$$G_K = \begin{bmatrix} G_0 + G_1 + G_3 & -G_3 \\ -G_3 & G_2 + G_3 \end{bmatrix}$$

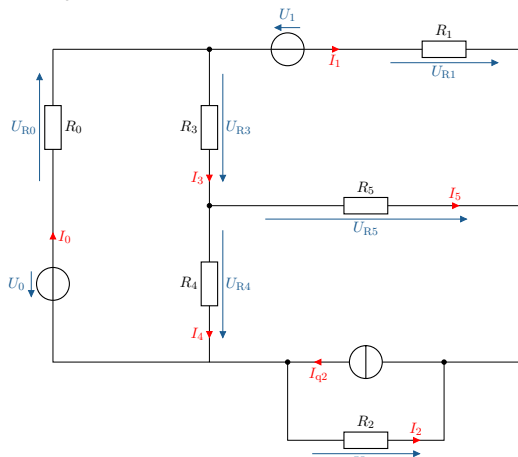
- Solving the system of equations using Gauss's method or Cramer's rule

$$\begin{bmatrix} G_0 + G_1 + G_3 & -G_3 \\ -G_3 & G_2 + G_3 \end{bmatrix} \cdot \begin{bmatrix} U_{K1-K0} \\ U_{K2-K0} \end{bmatrix} = \begin{bmatrix} -I_{q0} \\ -I_{q2} \end{bmatrix}$$

Example: Nodal analysis

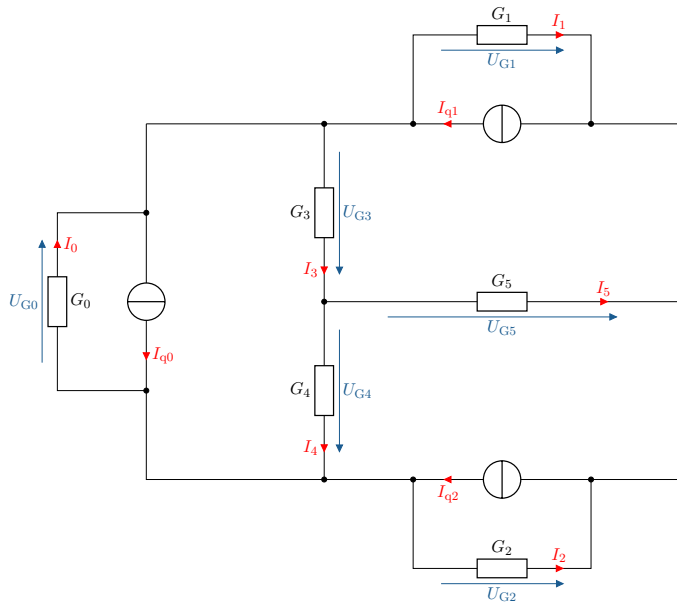
The following steps should be carried out for the node potential method:

- Preparation of the network
- Determination of nodes and node potentials
- Allocation of source currents
- Preparation of the conductivity matrix
- Set up a system of equations



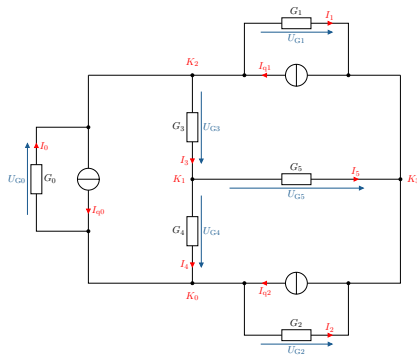
Example: Nodal analysis

a) Preparing the network:



Example: Nodal analysis

b) Determination of nodes and node potentials:



$$U_K = \begin{bmatrix} U_{K1-K0} \\ U_{K2-K0} \\ U_{K3-K0} \end{bmatrix}$$

c) Allocation of source currents:

$$I_K = \begin{bmatrix} 0 \\ -I_{q0} + I_{q1} \\ -I_{q1} - I_{q2} \end{bmatrix}$$

d) Determination of the conductance matrix:

$$G_K = \begin{bmatrix} G_3 + G_4 + G_5 & -G_3 & -G_5 \\ -G_3 & G_0 + G_1 + G_3 & -G_1 \\ -G_5 & -G_1 & G_1 + G_2 + G_5 \end{bmatrix}$$

e) Set up a system of equations:

$$\begin{bmatrix} G_3 + G_4 + G_5 & -G_3 & -G_5 \\ -G_3 & G_0 + G_1 + G_3 & -G_1 \\ -G_5 & -G_1 & G_1 + G_2 + G_5 \end{bmatrix} \cdot \begin{bmatrix} U_{K1-K0} \\ U_{K2-K0} \\ U_{K3-K0} \end{bmatrix} = \begin{bmatrix} 0 \\ -I_{q0} + I_{q1} \\ -I_{q1} - I_{q2} \end{bmatrix}$$

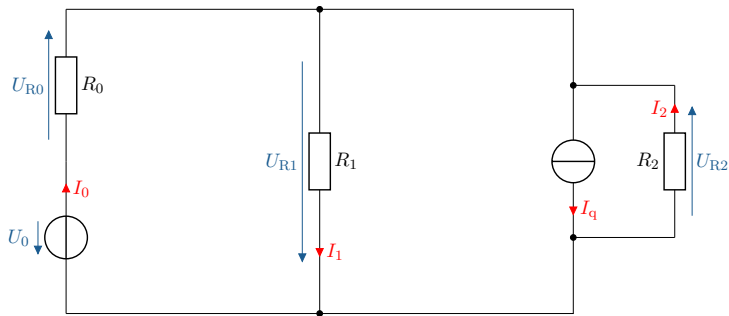
Key point: mesh flow method

The Students

- ▶ are familiar with the steps of the mesh flow process.
- ▶ can analyse electrical networks using the mesh current method.

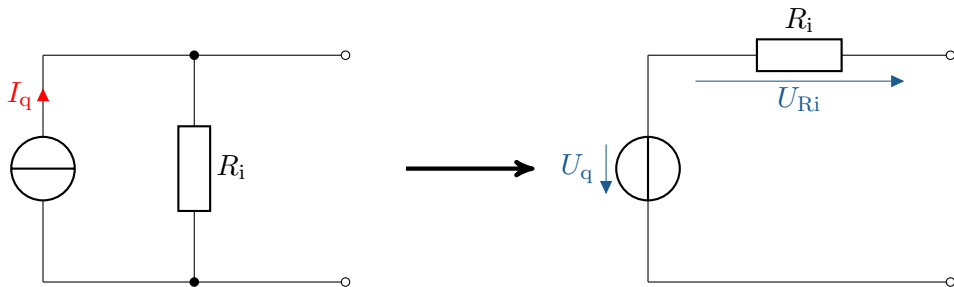
Mesh flow method

- ▶ Preparation of the network
- ▶ Define meshes
- ▶ Determine resistance matrix
- ▶ Assign source voltages
- ▶ set up a system of equations



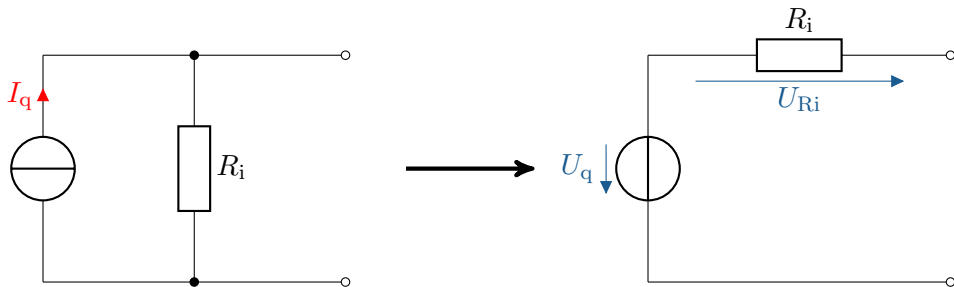
Preparation of the network

- Conversion of current sources into voltage sources:



Preparation of the network

- Conversion of current sources into voltage sources:

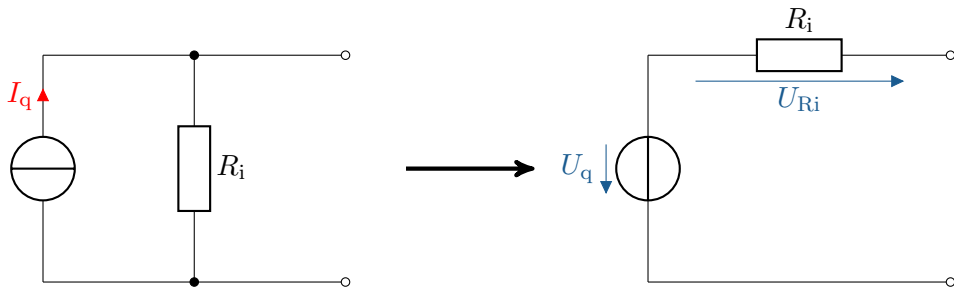


- Conversion of current values to voltage values:

$$U_q = R_i \cdot I_q$$

Preparation of the network

- Conversion of current sources into voltage sources:



- Conversion of current values to voltage values:

$$U_q = R_i \cdot I_q$$

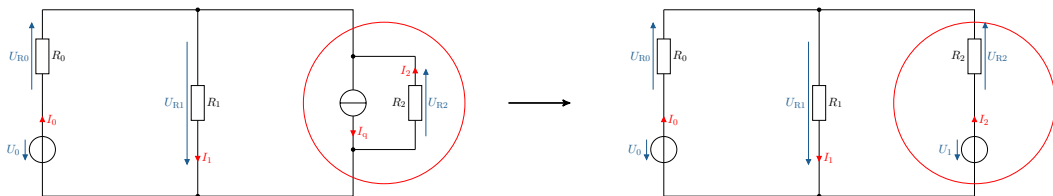
- Calculation of conductance values:

$$R_i = \frac{1}{G_i}$$

Preparation of the network

Conversion of the current source into a voltage source:

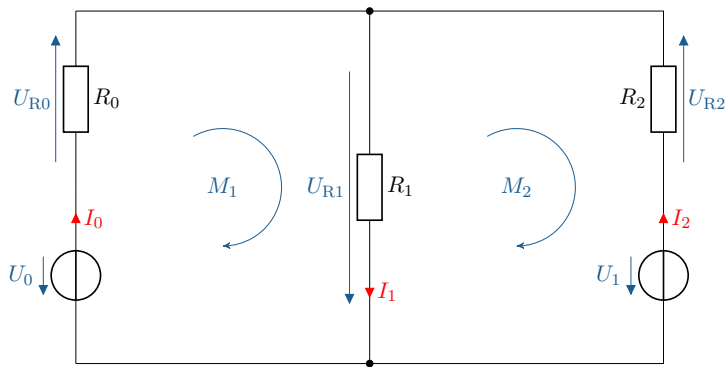
- Replace the power source with a voltage source
- The parallel resistor becomes a series resistor.



Key point: Mesh flow method

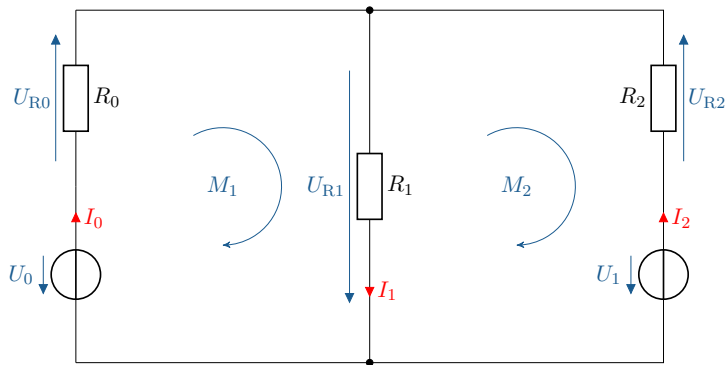
For the mesh current method, voltage sources and resistance values are required to analyse an electrical network.

► Determination of meshes and mesh flows



$$I_M = \begin{bmatrix} I_{M1} \\ I_{M2} \end{bmatrix}$$

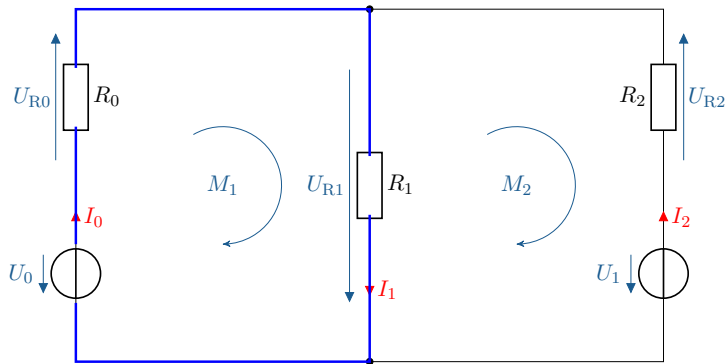
Determine resistance matrix



$$R_M = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Determine resistance matrix

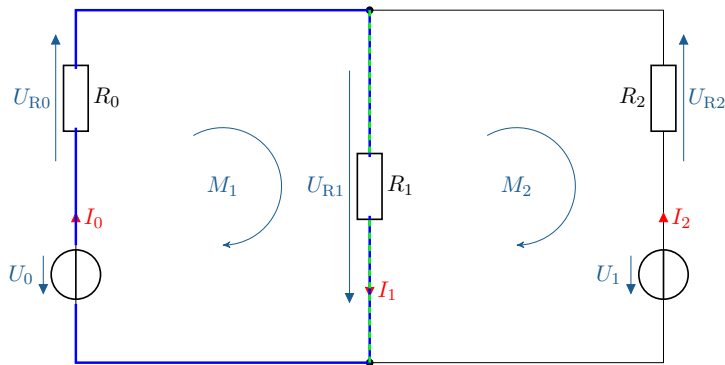
- Main diagonal with the resistance values directly adjacent to the meshes



$$R_M = \begin{bmatrix} R_0 + R_1 & 0 \\ 0 & R_1 + R_2 \end{bmatrix}$$

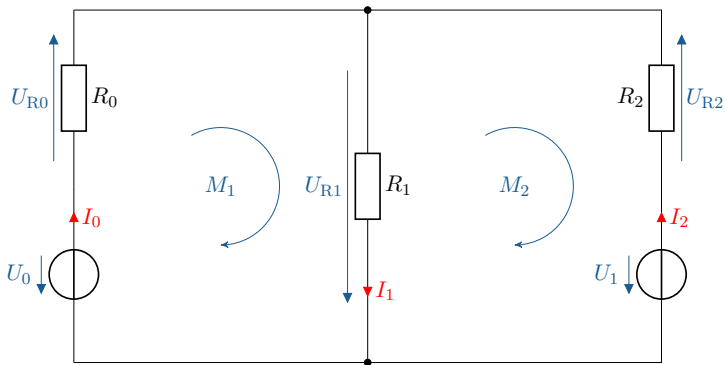
Determine resistance matrix

- ▶ Main diagonal with the resistance values directly adjacent to the meshes
- ▶ Other elements with resistance values that lie directly between the meshes involved



$$R_M = \begin{bmatrix} R_0 + R_1 & -R_1 \\ -R_1 & R_1 + R_2 \end{bmatrix}$$

Assign source voltages



$$U_M = \begin{bmatrix} U_0 \\ -U_1 \end{bmatrix}$$

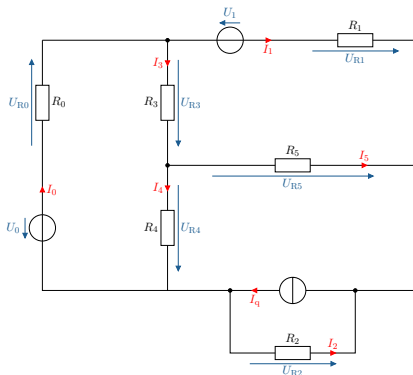
Set up a system of equations

$$\begin{bmatrix} R_0 + R_1 & -R_1 \\ -R_1 & R_1 + R_2 \end{bmatrix} \cdot \begin{bmatrix} I_{M1} \\ I_{M2} \end{bmatrix} = \begin{bmatrix} U_0 \\ -U_1 \end{bmatrix}$$

Example: Mesh flow method

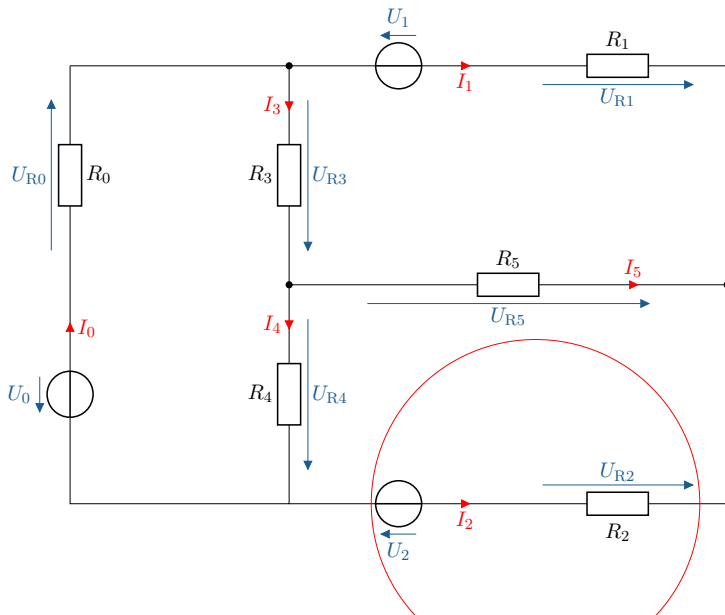
The following steps should be carried out for mesh flow analysis:

- Preparation of the network
- Define meshes and mesh flows
- Determine resistance matrix
- Assign source voltages
- Set up a system of equations



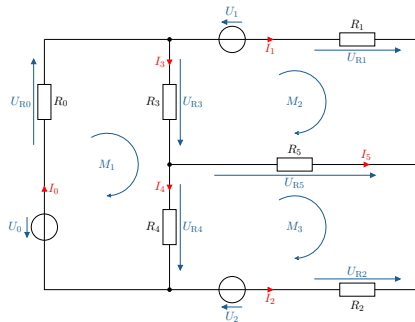
Example: Mesh flow method

a) Preparation of the network



Example: Mesh flow method

b) Define meshes and mesh flows



$$I_M = \begin{bmatrix} I_{M1} \\ I_{M2} \\ I_{M3} \end{bmatrix}$$

c) Determine resistance matrix

$$R_M = \begin{bmatrix} R_0 + R_3 + R_4 & -R_3 & -R_4 \\ -R_3 & R_1 + R_3 + R_5 & -R_5 \\ -R_4 & -R_5 & R_2 + R_4 + R_5 \end{bmatrix}$$

d) Assign source voltages

$$U_M = \begin{bmatrix} U_0 \\ U_1 \\ -U_2 \end{bmatrix}$$

e) Set up a system of equations

$$\begin{bmatrix} R_0 + R_3 + R_4 & -R_3 & -R_4 \\ -R_3 & R_1 + R_3 + R_5 & -R_5 \\ -R_4 & -R_5 & R_2 + R_4 + R_5 \end{bmatrix} \cdot \begin{bmatrix} I_{M1} \\ I_{M2} \\ I_{M3} \end{bmatrix} = \begin{bmatrix} U_0 \\ U_1 \\ -U_2 \end{bmatrix}$$

Calculate currents:

$$I_3 = I_0 - I_1$$

$$I_4 = I_3 - I_5$$

$$I_0 = I_4 - I_2$$

$$-I_5 = I_2 + I_1$$